Mugilan T

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2023	6.14
THE VELAMMAL INT SCH PANCHETTI THIRUVALLUR TN	Class XII (CBSE)	2023	89.40%
VELAMMAL MATRIC HR SEC SCHHOL PANCHETTY	Class X (SSLC)	2021	100.00%

SKILLS

Programming Languages: C, C++, Python

Web: HTML, CSS, JavaScript

Tools; VS Code, Git, Linux

RELEVANT COURSES

Data Structure and Applications: Learned various types of data structures, including arrays, linked lists, stacks, queues, trees, and graphs. Gained an understanding of their operations, properties, applications, and time and space complexity.

Operating System: Gained understanding of core OS concepts such as process scheduling, memory management (paging, segmentation, virtual memory), and synchronisation mechanisms using semaphores and locks.

Computer-Aided Numerical Methods: Applied numerical techniques, including Euler's method, Runge-Kutta (RK-4), and finite difference methods, to solve differential equations.

Machine Learning by Andrew Ng on Coursera : Gained foundational understanding of supervised and unsupervised learning techniques. Implemented algorithms such as linear regression, logistic regression, neural networks, and k-means clustering. Learned core concepts like cost functions, gradient descent, and regularisation.

Fundamentals of Machine Learning (FOML): Developed a strong **mathematical foundation** in machine learning, covering regression, classification, K-Nearest Neighbours (KNN), neural networks, and Graph Neural Networks (GNN). Studied the mathematical formulations, optimisation methods, and theoretical principles underlying machine learning models.

PROJECTS

Multithreaded Sudoku Validator (C, POSIX Threads):

- Implemented a Multithreaded Sudoku Validator in C, utilising POSIX Threads to concurrently check the validity of all nine rows, nine columns, and nine 3x3 subgrids.
- Managed thread creation and synchronisation to ensure correct and efficient task distribution across the grid

Restaurant Profit Predictor using Linear Regression (Python, NumPy, Matplotlib)

- Developed a Restaurant Profit Predictor by implementing a Linear Regression model from scratch in Python
- Applied fundamental Machine Learning concepts, including cost function calculation and gradient descent, to train the model and minimise prediction errors

Custom Machine Learning Library (Python, NumPy):

- Developed a custom machine learning library in Python with core modules for preprocessing, linear models, model selection, Naive Bayes, and neural networks.
- Implemented and evaluated Logistic Regression and other ML models on the Spambase dataset, including feature preprocessing, standardisation, and cross-validation.
- Built an autograd engine and neural network components with computational graph visualisation, enabling

gradient-based optimisation and model training.

Memory Match Game (HTML, CSS, JavaScript)

- Built a fully functional, browser-based Memory Match Game using HTML, CSS, and JavaScript
- Implemented card randomisation, pair-matching detection, dynamic card flipping, win/lose pop-ups and a real-time timer.

EXTRACURRICULAR

Dance: Interested in dancing and regularly practice it as a hobby.

Cricket: Passionate about cricket and regularly participates in informal matches.

Karate: Participated in District-Level Karate Competitions and secured Gold and Silver medals.